TUTORIAL 5

Matrix Operations, Bases, and Dimension

MOTIVATION AND GUIDING IDEA

Motivation. Matrix multiplication describes composition of linear operations: in ABx, apply B first and then A. Checking dimensions reveals in advance whether the composition is possible. A basis gives a unique description of every vector, and dimension counts the coefficients required for that description.

NOTATION AND CONVENTIONS

Notation guide. $M_{m\times n}(\mathbb{F})$ is the space of matrices with m rows and n columns; a_{ij} is the entry in row i, column j. $I_n = (\delta_{ij})$, where $\delta_{ij} = 1$ if $i = j$ and is zero otherwise. e_j is the jth standard basis vector, and Ae_j is column j of A. A^T denotes transpose, $\mathrm{diag}(d_1, \ldots, d_n)$ a diagonal matrix, $|B|$ the cardinality of a finite set, and $\dim_{\mathbb{F}} V$ the dimension over $\mathbb{F}$.

DEFINITION 5.1

Hypotheses to check. If $A \in M_{k\times m}(\mathbb{F})$ and $B \in M_{m\times n}(\mathbb{F})$, then $AB \in M_{k\times n}(\mathbb{F})$ with $(AB)_{ij} = \sum_{r=1}^{m} a_{ir}b_{rj}$. Also $(AB)^T = B^T A^T$. Over $\mathbb{C}$, the transpose A^T does not include complex conjugation.

For square matrices, define $\mathrm{Sym}_n(\mathbb{F}) = \{A : A^T = A\}$ and $\mathrm{ASym}_n(\mathbb{F}) = \{A : A^T = -A\}$. Both sets are subspaces over any field. If the characteristic is different from 2, a skew-symmetric matrix has zero diagonal and every matrix has a unique symmetric/skew-symmetric decomposition using division by 2. In characteristic 2 the two conditions coincide and do not imply a zero diagonal.

If $\dim_{\mathbb{F}} V = n < \infty$, a set of n vectors is a basis if and only if it is independent, and if and only if it spans. For $U, W \le V$, $\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$; subtracting the intersection dimension prevents shared directions from being counted twice.

MOTIVATION AND GUIDING IDEA

Tutorial objectives. Compute with matrices while checking dimensions and multiplication order, recognize symmetric structure, and choose bases. Connect algebraic calculations with sums and intersections of subspaces.

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5.1 Matrix Arithmetic

Throughout this section, let

$$A = \begin{pmatrix} 1 & 5 \\ 2 & 3 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 1 \\ 3 & 6 \end{pmatrix}.$$

5.1.1 Scalar Multiplication

Scalar multiplication is defined for every matrix. Multiply **every** entry by the scalar: $(\lambda A)_{ij} = \lambda(A)_{ij}$.

EXAMPLE 5.1

Compute

$$7A = \begin{pmatrix} 7\cdot 1 & 7\cdot 5 \\ 7\cdot 2 & 7\cdot 3 \end{pmatrix} = \begin{pmatrix} 7 & 35 \\ 14 & 21 \end{pmatrix}.$$

5.1.2 Matrix Addition and Subtraction

These are defined only for two matrices of **exactly the same shape**. Add or subtract corresponding entries: $(A \pm B)_{ij} = (A)_{ij} \pm (B)_{ij}$.

EXAMPLE 5.2

Compute

$$A + B = \begin{pmatrix} 1 & 5 \\ 2 & 3 \end{pmatrix} + \begin{pmatrix} 1 & 1 \\ 3 & 6 \end{pmatrix} = \begin{pmatrix} 1+1 & 5+1 \\ 2+3 & 3+6 \end{pmatrix} = \begin{pmatrix} 2 & 6 \\ 5 & 9 \end{pmatrix}.$$

5.1.3 Matrix Multiplication

A product is defined only when the number of columns of the first (left) matrix equals the number of rows of the second (right) matrix. If A is $k \times m$ and B is $m \times n$, then AB is $k \times n$ and is defined by

$$(AB)_{ij} = \sum_{r=1}^{m} a_{ir} b_{rj} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{im}b_{mj}.$$

This is the entry in row i, column j of the product. Follow row i of A and column j of B, each containing m entries: multiply matching entries and add. **Note:** the existence of AB does **not** ensure the existence of BA. With the dimensions above, the latter also requires $n = k$.

EXERCISE 5.1

Please compute AB and BA for the matrices defined above.

WORKED SOLUTION

$$AB = \begin{pmatrix} 1\cdot 1 + 5\cdot 3 & 1\cdot 1 + 5\cdot 6 \\ 2\cdot 1 + 3\cdot 3 & 2\cdot 1 + 3\cdot 6 \end{pmatrix} = \begin{pmatrix} 16 & 31 \\ 11 & 20 \end{pmatrix},$$

$$BA = \begin{pmatrix} 1\cdot 1 + 1\cdot 2 & 1\cdot 5 + 1\cdot 3 \\ 3\cdot 1 + 6\cdot 2 & 3\cdot 5 + 6\cdot 3 \end{pmatrix} = \begin{pmatrix} 3 & 8 \\ 15 & 33 \end{pmatrix}.$$

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EXERCISE 5.1 CONTINUED

For each entry we added the products of matching row and column entries. The results differ because the rows and columns involved change with the order of the factors.

NOTE

Important conclusion: matrix multiplication is not commutative. The equality $AB = BA$ need not hold. In the example, $AB \neq BA$ even though both products are defined.

EXERCISE 5.2

Let A be a $k \times m$ matrix with a zero row, and let B be $m \times n$. Please prove that the $k \times n$ matrix AB has a zero row.

WORKED SOLUTION

Let i be the index of a zero row of A, where $1 \leq i \leq k$. Thus $(A)_{ir} = 0$ for every $1 \leq r \leq m$. In row i of AB, for every $1 \leq j \leq n$,

$$(AB)_{ij} = \sum_{r=1}^{m} (A)_{ir}(B)_{rj} = \sum_{r=1}^{m} 0 = 0.$$

All entries in row i of AB are zero, as required. **The zero matrix** is a matrix of any specified shape whose entries are all zero.

5.2 Diagonal Matrices

A **diagonal matrix** is a **square** matrix whose off-diagonal entries are all zero. Write

$$D = \operatorname{diag}(d_1, \ldots, d_n) = \begin{pmatrix} d_1 & & 0 \\ & \ddots & \\ 0 & & d_n \end{pmatrix}.$$

Equivalently,

$$[D]_{ij} = \begin{cases} 0, & i \neq j, \\ d_i, & i = j. \end{cases}$$

A **scalar matrix** is a diagonal matrix with equal diagonal entries. The **identity matrix** is a diagonal matrix with all diagonal entries equal to 1, hence is a scalar matrix. A scalar matrix can therefore be written aI, where a is its common diagonal entry. Write I_n for the $n \times n$ identity, or simply I when its size is clear. For every $n \times m$ matrix A, $AI_m = I_n A = A$.

EXAMPLE 5.3

$$IA = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 5 \\ 2 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 5 \\ 2 & 3 \end{pmatrix} = A, \qquad AI = \begin{pmatrix} 1 & 5 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 5 \\ 2 & 3 \end{pmatrix}.$$

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EXERCISE 5.3

Let A, B be diagonal matrices of order n. Please prove that:

1. $A + B$ is diagonal.
2. AB is diagonal and $[AB]_{ii} = a_{ii}b_{ii}$ for every $1 \leq i \leq n$.
3. A and B **commute**: $AB = BA$.

WORKED SOLUTION

1. For $i \neq j$, $(A + B)_{ij} = a_{ij} + b_{ij} = 0 + 0 = 0$. Thus $A + B$ is diagonal, with diagonal entries $(A + B)_{ii} = a_{ii} + b_{ii}$.
2. By the definition of multiplication,

$$(AB)_{ij} = \sum_{k=1}^{n} a_{ik}b_{kj}.$$

   In row i of A, only a_{ii} can be nonzero, so the sum reduces to $a_{ii}b_{ij}$. If $i \neq j$, then $b_{ij} = 0$, giving $(AB)_{ij} = 0$. If $i = j$, then $(AB)_{ii} = a_{ii}b_{ii}$. Hence AB is diagonal.
3. Similarly BA is diagonal with diagonal entries $b_{ii}a_{ii}$. Scalars in a field commute, so

$$a_{ii}b_{ii} = b_{ii}a_{ii} \qquad (1 \leq i \leq n).$$

   All entries of AB and BA agree; therefore $AB = BA$.

NOTE

Check dimensions before computing. The product $A_{k\times m}B_{m\times n}$ has k rows and n columns. Equality of matrices requires both equal dimensions and equal corresponding entries.

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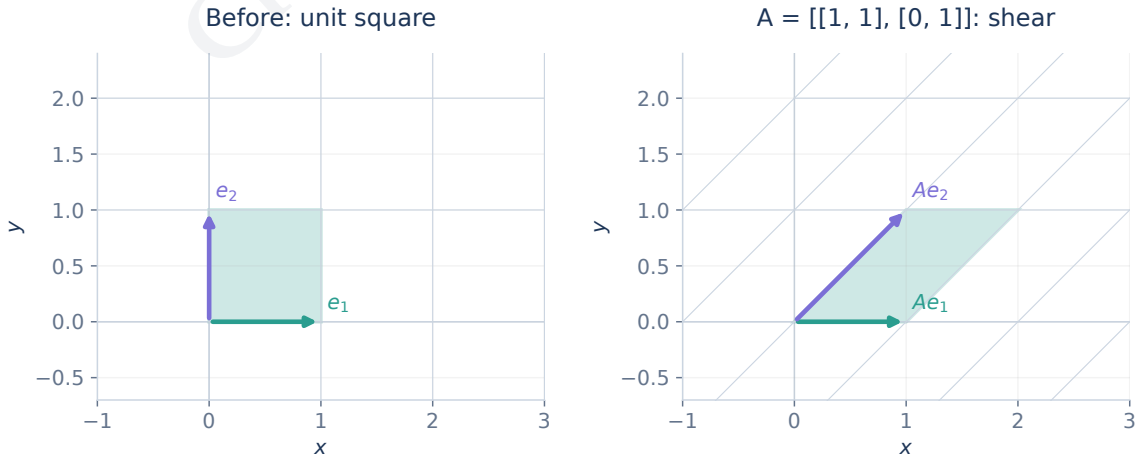


Figure 5.1. A linear map takes a straight grid to a straight grid. The columns of the matrix describe the images of the basis vectors.

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5.3 Transpose

- The transpose is defined for every matrix. The **transpose** A^T satisfies $(A^T)_{ij} = (A)_{ji}$.
- Rows become columns and columns become rows.
- If $A \in \mathbb{R}^{m\times n}$, then $A^T \in \mathbb{R}^{n\times m}$: the numbers of rows and columns are interchanged.

EXAMPLE 5.4

If

$$C = \begin{pmatrix} 1 & -2 \\ 4 & 5 \\ 7 & 3 \end{pmatrix} \in \mathbb{R}^{3\times 2}, \qquad C^T = \begin{pmatrix} 1 & 4 & 7 \\ -2 & 5 & 3 \end{pmatrix} \in \mathbb{R}^{2\times 3}.$$

The matrix C has 3 rows and 2 columns. Transposing gives a matrix with 3 columns and 2 rows.

5.4 Symmetry and Skew-Symmetry

- A **square** matrix A is **symmetric** if $A^T = A$.
- A **square** matrix A is **skew-symmetric** if $A^T = -A$.
- Every diagonal matrix is symmetric; in particular, every scalar matrix and the identity matrix are symmetric.

The square condition matters: $A \in \mathbb{F}^{m\times n}$ and $A^T \in \mathbb{F}^{n\times m}$ have the same shape only when $n = m$. Intuitively, symmetry means reflection across the main diagonal; skew-symmetry means the same reflection followed by a change of sign.

EXERCISE 5.4

Over $\mathbb{R}$, please determine whether each of the following matrices is symmetric, skew-symmetric, or neither:

$$A = \begin{pmatrix} 1 & 3 \\ 3 & 2 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}, \qquad C = \begin{pmatrix} 1 & \pi \\ -5 & 0 \end{pmatrix}.$$

WORKED SOLUTION

1. $A = \begin{pmatrix} 1 & 3 \\ 3 & 2 \end{pmatrix}$ has $A^T = \begin{pmatrix} 1 & 3 \\ 3 & 2 \end{pmatrix}$. Thus $A^T = A$ and A is symmetric.
2. $B = \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}$ has $B^T = \begin{pmatrix} 0 & 2 \\ -2 & 0 \end{pmatrix}$. Thus $B^T = -B$ and B is skew-symmetric.
3. $C = \begin{pmatrix} 1 & \pi \\ -5 & 0 \end{pmatrix}$ has $C^T = \begin{pmatrix} 1 & -5 \\ \pi & 0 \end{pmatrix}$. This equals neither C nor $-C$, so C is neither symmetric nor skew-symmetric.

As discussed in class, sums and scalar multiples of symmetric matrices are symmetric. This assertion also appears in the exercise later in this chapter.

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EXERCISE 5.5

Please prove or disprove.

1. If A, B are symmetric, then AB is symmetric.
2. If A, B are both nonsymmetric, then AB is nonsymmetric.

WORKED SOLUTION

Part 1. Investigate the transpose:

$$(AB)^T = B^T A^T = BA.$$

The first equality is the transpose-of-a-product rule; the second uses symmetry. Since BA need not equal AB, consider the counterexample

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 2 \\ 2 & 0 \end{pmatrix}, \quad AB = \begin{pmatrix} 3 & 2 \\ 1 & 2 \end{pmatrix}.$$

Both factors are symmetric, but their product is not.

Part 2. Take $A = B = \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix}$. Neither is symmetric, but

$$AB = \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}.$$

The product is diagonal and hence symmetric, disproving the assertion.

EXERCISE 5.6

Let $A \in M_n(\mathbb{F})$, where $\text{char}(\mathbb{F}) \neq 2$. Please prove that:

1. $A + A^T$ is symmetric.
2. $A - A^T$ is skew-symmetric.
3. A can be written as the sum of a symmetric matrix and a skew-symmetric matrix.

WORKED SOLUTION

1. $(A + A^T)^T = A^T + (A^T)^T = A^T + A = A + A^T$.
2. $(A - A^T)^T = A^T - A = -(A - A^T)$, proving skew-symmetry.
3. Add the matrices from the preceding parts:

$$2A = (A + A^T) + (A - A^T),$$
$$A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T).$$

As shown above, the first summand is symmetric and the second is skew-symmetric.

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5.5 Intersections and Sums of Subspaces

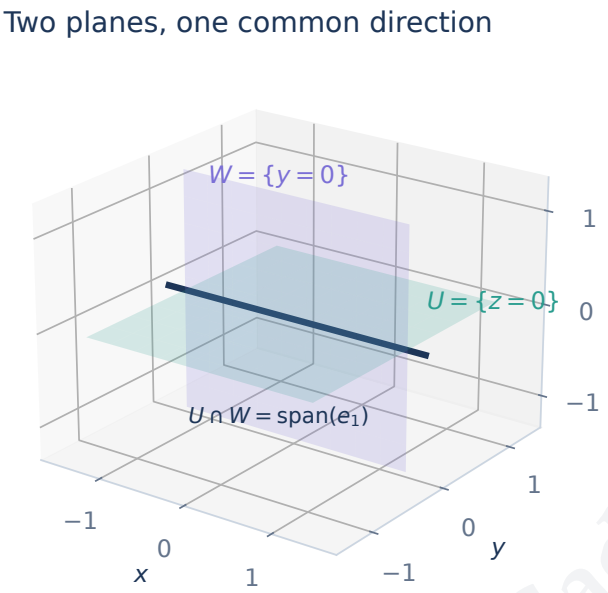


Figure 5.2. Two distinct planes through the origin in $\mathbb{R}^3$ may intersect in a line. Here $U = \{z = 0\}$ and $W = \{y = 0\}$, so $U \cap W = \operatorname{span}\{(1, 0, 0)\}$ and $U + W = \mathbb{R}^3$.

RECALL AND TOOLS

For subspaces $U, W \subseteq V$ of a vector space V, their intersection is

$$U \cap W = \{v \in V : v \in U \text{ and } v \in W\}.$$

DEFINITION 5.2

For subspaces $U, W \subseteq V$, define

$$U + W = \{u + w : u \in U, w \in W\}.$$

This is a subspace of V containing U and W. If also $U \cap W = \{0\}$, the sum is a **direct sum**, denoted $U \oplus W$.

EXERCISE 5.7

Let

$$\operatorname{Sym}(\mathbb{F}) = \{A \in M_n(\mathbb{F}) : A = A^T\}, \qquad \operatorname{aSym}(\mathbb{F}) = \{A \in M_n(\mathbb{F}) : A = -A^T\}.$$

Please prove that the symmetric and skew-symmetric matrices are subspaces of $M_n(\mathbb{F})$, and that whenever $\operatorname{char}(\mathbb{F}) \neq 2$,

$$M_n(\mathbb{F}) = \operatorname{Sym}(\mathbb{F}) \oplus \operatorname{aSym}(\mathbb{F}).$$

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EXERCISE 5.7 CONTINUED

WORKED SOLUTION

We show that $\mathrm{Sym}(\mathbb{F})$ is a subspace. For the skew-symmetric set, the corresponding identities are

$$0^T = -0, \quad (A+B)^T = -A - B = -(A+B), \quad (\lambda A)^T = -\lambda A,$$

so this set is also a subspace. For the symmetric set, take $A, B \in \mathrm{Sym}(\mathbb{F})$ and $\lambda \in \mathbb{F}$. Clearly $0 \in \mathrm{Sym}(\mathbb{F})$. Closure under addition follows from $(A+B)^T = A^T + B^T = A + B$. Closure under scalar multiplication follows from $(\lambda A)^T = \lambda A^T = \lambda A$. To establish the direct sum, first prove equality with the sum and then show that the intersection is zero. Check both inclusions for $M_n(\mathbb{F}) = \mathrm{Sym}(\mathbb{F}) + \mathrm{aSym}(\mathbb{F})$:

1. $M_n(\mathbb{F}) \supseteq \mathrm{Sym}(\mathbb{F}) + \mathrm{aSym}(\mathbb{F})$ follows because both summands are subspaces of $M_n(\mathbb{F})$.
2. Take $A \in M_n(\mathbb{F})$. Then
$$A = \frac{A + A^T}{2} + \frac{A - A^T}{2}.$$
The first summand is symmetric because
$$\left(\frac{A + A^T}{2}\right)^T = \frac{A^T + (A^T)^T}{2} = \frac{A + A^T}{2}.$$
The second is skew-symmetric because
$$\left(\frac{A - A^T}{2}\right)^T = \frac{A^T - (A^T)^T}{2} = \frac{A^T - A}{2} = -\frac{A - A^T}{2}.$$
This proves the reverse inclusion.

Now take $A \in \mathrm{Sym}(\mathbb{F}) \cap \mathrm{aSym}(\mathbb{F})$. Symmetry gives $A = A^T$, and skew-symmetry gives $-A = A^T$. Therefore $A = -A$, so $a_{ij} = -a_{ij}$ for every $1 \le i, j \le n$. Thus $a_{ij} + a_{ij} = 2a_{ij} = 0$. As the field has characteristic different from 2, the scalar 2 is nonzero, so every $a_{ij} = 0$. Hence $A = 0$, as required.

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5.6 Bases and Dimension

DEFINITION 5.3

Let V be a vector space over a field $\mathbb{F}$. A set $\{u_1, \ldots, u_n\} \subseteq V$ is a **basis** of V if it is linearly independent and spans V.

A finite-dimensional vector space can have many bases, but all its bases have the same number of elements. This number is the **dimension** of the space.

EXAMPLE 5.5

$$\dim(\mathbb{F}^n) = n, \qquad \dim(M_{n\times m}(\mathbb{F})) = nm, \qquad \dim(\mathbb{F}_n[x]) = n+1.$$

Familiar spaces have **standard bases**, denoted $e_1, e_2, \ldots, e_{\dim V}$. For instance, in $\mathbb{R}^2$, $e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

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and $e_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$; in $\mathbb{R}_2[x]$, $e_1 = 1, e_2 = x, e_3 = x^2$.

RECALL AND TOOLS

In a vector space of known finite dimension n, any two of the following three properties suffice to establish that a set is a basis:

1. Linear independence.
2. Spanning the space.
3. Having exactly n elements, the dimension of the space.

EXERCISE 5.8

Please determine whether

$$v_1 = (1, 0, i)^T, \qquad v_2 = (1 + i, 1, 0)^T, \qquad v_3 = (2 + i, 1 + i, 1 + 2i)^T$$

form a basis of $\mathbb{C}^3$. If not, please give an explicit dependence relation.

WORKED SOLUTION

For three vectors in $\mathbb{C}^3$, it suffices to test independence. Place them in columns and row-reduce:

$$\begin{pmatrix} 1 & 1+i & 2+i \\ 0 & 1 & 1+i \\ i & 0 & 1+2i \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - iR_1} \begin{pmatrix} 1 & 1+i & 2+i \\ 0 & 1 & 1+i \\ 0 & 1-i & 2 \end{pmatrix}$$

$$\xrightarrow{R_3 \leftarrow R_3 - (1-i)R_2} \begin{pmatrix} 1 & 1+i & 2+i \\ 0 & 1 & 1+i \\ 0 & 0 & 0 \end{pmatrix}.$$

The last step uses $(1 - i)(1 + i) = 2$. The rank is 2, so the set is not a basis. To identify the relation, verify directly:

$$(2 - i)v_1 + (1 + i)v_2 = \begin{pmatrix} (2 - i) + (1 + i)^2 \\ 1 + i \\ (2 - i)i \end{pmatrix} = \begin{pmatrix} 2 + i \\ 1 + i \\ 1 + 2i \end{pmatrix} = v_3.$$

Thus $(2 - i)v_1 + (1 + i)v_2 - v_3 = 0$ is a nontrivial relation confirming the row reduction.

EXERCISE 5.9

Please determine whether $B = (x^2 - 1, 2x + 3, 1 + x + x^2)$ is a basis of $\mathbb{R}_2[x]$, and then whether it is a basis of $\mathbb{R}_3[x]$. If it is a basis in the first case, please find the coordinates of $p(x) = ax^2 + bx + c$.

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EXERCISE 5.9 CONTINUED

WORKED SOLUTION

Our convention is that $\mathbb{R}_j[x]$ consists of polynomials of degree at most j. Seek α, β, γ such that

$$ax^2 + bx + c = \alpha(x^2 - 1) + \beta(2x + 3) + \gamma(1 + x + x^2).$$

Comparing coefficients gives

$$\alpha + \gamma = a, \qquad 2\beta + \gamma = b, \qquad -\alpha + 3\beta + \gamma = c.$$

The second equation gives $\gamma = b - 2\beta$, and the first gives $\alpha = a - b + 2\beta$. Substitute in the third:

$$-(a - b + 2\beta) + 3\beta + b - 2\beta = c \iff -a + 2b - \beta = c.$$

Consequently,

$$\beta = -a + 2b - c, \quad \gamma = 2a - 3b + 2c, \quad \alpha = -a + 3b - 2c.$$

For every a, b, c there is a unique solution, so B is a basis of $\mathbb{R}_2[x]$, with

$$[p]_B = (-a + 3b - 2c,\ -a + 2b - c,\ 2a - 3b + 2c)^T.$$

However, B does not span $\mathbb{R}_3[x]$: every linear combination of its members has degree at most 2, and therefore cannot equal x^3.

EXERCISE 5.10

Let $\{u, v, w\}$ be a basis of $\mathbb{R}^3$. Are the following sets also bases of $\mathbb{R}^3$?

1. $\{u, u + v, u - v\}$.
2. $\{\frac{1}{2}u, \frac{1}{3}v, \frac{1}{4}w\}$.
3. $\{u + v + w, u + 2v, v - w\}$.

WORKED SOLUTION

Each displayed collection has three vectors in a space of dimension 3, so testing independence determines whether it is a basis.

1. Suppose $\alpha_1 u + \alpha_2(u + v) + \alpha_3(u - v) = 0$. Rearrange:

   $$(\alpha_1 + \alpha_2 + \alpha_3)u + (\alpha_2 - \alpha_3)v + 0 \cdot w = 0.$$

   Since $\{u, v, w\}$ is a basis, independence forces all the rearranged coefficients to be zero:

   $$\begin{cases} \alpha_1 + \alpha_2 + \alpha_3 = 0, \\ \alpha_2 - \alpha_3 = 0. \end{cases}$$

   There are nontrivial solutions: $\alpha_2 = \alpha_3$ and $\alpha_1 = -2\alpha_3$. Thus the set is dependent and is not a basis.

2. Suppose $\alpha_1 \frac{1}{2}u + \alpha_2 \frac{1}{3}v + \alpha_3 \frac{1}{4}w = 0$. Since $\{u, v, w\}$ is a basis, $\frac{\alpha_1}{2} = 0$, $\frac{\alpha_2}{3} = 0$, and $\frac{\alpha_3}{4} = 0$. Hence the set is independent and is a basis.

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EXERCISE 5.10 CONTINUED

3. Suppose $\alpha_1(u+v+w)+\alpha_2(u+2v)+\alpha_3(v-w)=0$. Rearrange:

$$(\alpha_1+\alpha_2)u+(\alpha_1+2\alpha_2+\alpha_3)v+(\alpha_1-\alpha_3)w=0.$$

The basis property gives

$$\begin{cases}\alpha_1+\alpha_2=0,\\ \alpha_1+2\alpha_2+\alpha_3=0,\\ \alpha_1-\alpha_3=0.\end{cases}$$

In matrix form,

$$\begin{pmatrix}1&1&0\\1&2&1\\1&0&-1\end{pmatrix}\dashrightarrow\begin{pmatrix}1&0&-1\\0&1&1\\0&0&0\end{pmatrix}.$$

There are solutions other than the trivial solution, so the set is dependent and is not a basis.

5.7 The Dimension Formula

PROPOSITION 5.1

Let V be a finitely generated vector space over a field $\mathbb{F}$, and let U be a subspace of V.

1. U is finitely generated and $\dim U \le \dim V$.
2. $\dim U = \dim V$ if and only if $U = V$.

The dimension formula:

PROPOSITION 5.2

If V is finite-dimensional over $\mathbb{F}$ and $U, W \subseteq V$ are subspaces, then

$$\dim(U+W)=\dim U+\dim W-\dim(U\cap W).$$

EXERCISE 5.11

1. Let V have dimension n over $\mathbb{F}$, and let $U, W \subseteq V$ be subspaces such that $\dim U = n-1$ and $W \nsubseteq U$. Please prove that:
   (a) $V = U+W$.
   (b) $\dim(U\cap W)=\dim W-1$.
2. Let $V=\mathbb{R}_2[X]$, $U=\operatorname{span}\{X, X^2+1\}$, and $W=\{P\in V : P'(1)=0\}$.
   (a) Please compute $\dim(U+W)$ and $\dim(U\cap W)$.
   (b) Is $U+W=V$? If so, is the sum direct, that is, is $U\oplus W=V$?

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EXERCISE 5.11 CONTINUED

WORKED SOLUTION

1. (a) By assumption, there is $x \in W \setminus U$. Hence $U \subsetneq U + W \subseteq V$, and the preceding dimension lemma gives

$$n - 1 = \dim U < \dim(U + W) \leq \dim V = n.$$

Thus $\dim(U + W) = n$, and the equality case in the lemma gives $V = U + W$.

(b) Apply the dimension formula:

$$\begin{aligned} \dim(U + W) &= \dim U + \dim W - \dim(U \cap W), \\ n &= n - 1 + \dim W - \dim(U \cap W). \end{aligned}$$

Therefore $\dim(U \cap W) = \dim W - 1$, as required.

2. (a) We have $\dim V = 3$ and $\dim U = 2$, since its displayed spanning vectors are independent. Also $W \not\subseteq U$: for instance, $Q(X) = 1$ belongs to $W \setminus U$. All hypotheses of part 1 hold, giving $V = U + W$ and hence $\dim(U + W) = \dim V = 3$. The second conclusion of part 1 gives $\dim(U \cap W) = \dim W - 1$, so it remains to find $\dim W$. For $P(X) = aX^2 + bX + c \in W$, with $a, b, c \in \mathbb{R}$, the condition is $P'(1) = 2a + b = 0$, that is, $b = -2a$. Thus

$$\begin{aligned} W &= \{aX^2 - 2aX + c : a, c \in \mathbb{R}\} \\ &= \{a(X^2 - 2X) + c \cdot 1 : a, c \in \mathbb{R}\} \\ &= \operatorname{span}\{1, X^2 - 2X\}. \end{aligned}$$

Consequently $\dim W = 2$, and $\dim(U \cap W) = 1$.

(b) This is indeed a sum equal to V: $U + W \subseteq V$, and $\dim(U + W) = \dim V$ implies $U + W = V$. The sum is not direct, since

$$(X - 1)^2 = (X^2 + 1) - 2X \in U, \qquad \frac{d}{dX}(X - 1)^2 \Big|_{X=1} = 0,$$

so $(X - 1)^2 \in W$. In fact $U \cap W = \operatorname{span}\{(X - 1)^2\}$, consistent with the computed dimension 1.

EXERCISE 5.12

A drafting program applies a stretch and a shear in the plane:

$$D = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}, \qquad H = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}.$$

Please compute DH and HD entry by entry, please compare their action on e_2, and please find all v such that $DHv = HDv$.

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EXERCISE 5.12 CONTINUED

WORKED SOLUTION

By definition, each entry is a sum of row-by-column products:

$$DH = \begin{pmatrix} 3\cdot 1+0\cdot 0 & 3\cdot 2+0\cdot 1 \\ 0\cdot 1+1\cdot 0 & 0\cdot 2+1\cdot 1 \end{pmatrix} = \begin{pmatrix} 3 & 6 \\ 0 & 1 \end{pmatrix},$$

$$HD = \begin{pmatrix} 1\cdot 3+2\cdot 0 & 1\cdot 0+2\cdot 1 \\ 0\cdot 3+1\cdot 0 & 0\cdot 0+1\cdot 1 \end{pmatrix} = \begin{pmatrix} 3 & 2 \\ 0 & 1 \end{pmatrix}.$$

Therefore $DHe_2 = (6,1)^T$, whereas $HDe_2 = (2,1)^T$. The rightmost operation is applied first, and changing the order changes the result. For $v = (x,y)^T$, the difference between the results is $((3x+6y)-(3x+2y),\ y-y)^T = (4y,0)^T$. The results agree if and only if $y = 0$, precisely on $\operatorname{span}\{e_1\}$.

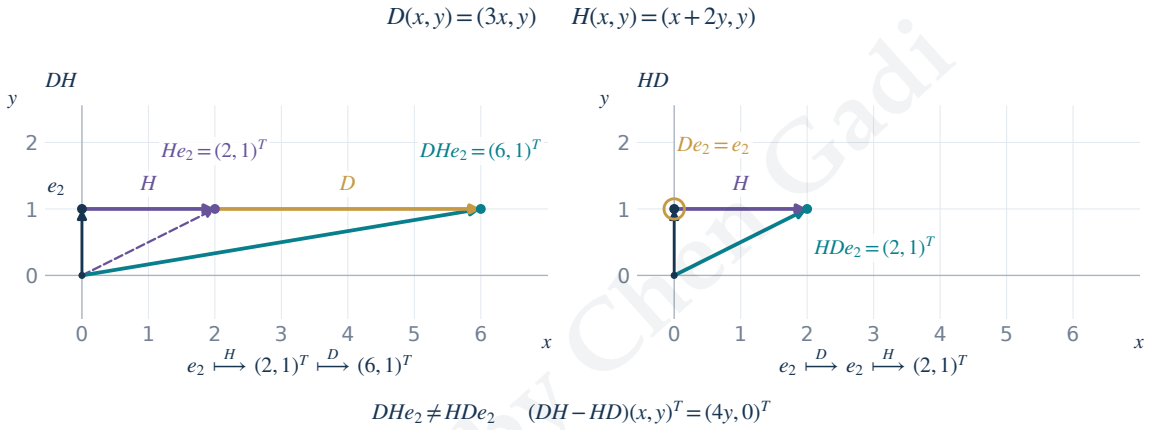


Figure 5.3. Operation order on the same vector: DH applies H first, whereas HD applies D first. The ring marks the fixed point $De_2 = e_2$, but the final results differ: $DHe_2 = (6,1)^T \neq (2,1)^T = HDe_2$.

Additional practice

EXERCISE 5.13 FOUNDATION

Over $\mathbb{R}$, let

$$A = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 2 \\ 3 & 0 \\ 0 & 1 \end{pmatrix}.$$

Please determine the dimensions of AB, BA, please compute both products, and please use $e_2 \in \mathbb{R}^2$ to verify $(AB)e_2 = A(Be_2)$. Can AB and BA be equal as matrices?

WORKED SOLUTION

The shapes of A, B are 2×3 and 3×2, so AB has order 2 and BA order 3. Entry-by-entry calculation gives

$$AB = \begin{pmatrix} 1\cdot 1+0\cdot 3+2\cdot 0 & 1\cdot 2+0\cdot 0+2\cdot 1 \\ 0\cdot 1+1\cdot 3-1\cdot 0 & 0\cdot 2+1\cdot 0-1\cdot 1 \end{pmatrix} = \begin{pmatrix} 1 & 4 \\ 3 & -1 \end{pmatrix},$$

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EXERCISE 5.13 FOUNDATION CONTINUED

$$BA = \begin{pmatrix} 1\cdot 1+2\cdot 0 & 1\cdot 0+2\cdot 1 & 1\cdot 2-2\cdot 1 \\ 3\cdot 1+0\cdot 0 & 3\cdot 0+0\cdot 1 & 3\cdot 2-0\cdot 1 \\ 0\cdot 1+1\cdot 0 & 0\cdot 0+1\cdot 1 & 0\cdot 2-1\cdot 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 0 \\ 3 & 0 & 6 \\ 0 & 1 & -1 \end{pmatrix}.$$

Now $Be_2 = (2,0,1)^T$, so $A(Be_2) = (4,-1)^T$. Also $(AB)e_2$ is the second column of AB, namely $(4,-1)^T$. Both products are defined, but they have different dimensions; they act on different spaces and cannot be equal as matrices.

EXERCISE 5.14 CHALLENGE

Let $n \geq 2$ and $A \in M_n(\mathbb{F})$ over an arbitrary field. Please prove:

1. A commutes with every diagonal matrix if and only if A is diagonal.
2. A commutes with every matrix in $M_n(\mathbb{F})$ if and only if $A = \lambda I_n$ for some $\lambda \in \mathbb{F}$.

You may use the matrix units E_{ij}, whose entry (i,j) is 1 and all other entries are zero.

WORKED SOLUTION

Suppose A commutes with every diagonal matrix. In particular, $AE_{kk} = E_{kk}A$ for every k. In entry (i,k) with $i \neq k$, the left side has value a_{ik} and the right side has value zero. Hence $a_{ik} = 0$ for every $i \neq k$, so A is diagonal. Conversely, two diagonal matrices commute because their product is diagonal and its diagonal entries are products of commuting scalars.
If A commutes with every matrix, part 1 gives $A = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$. For $i \neq j$,

$$AE_{ij} = \lambda_i E_{ij}, \qquad E_{ij}A = \lambda_j E_{ij}.$$

Comparing entry (i,j) gives $\lambda_i = \lambda_j$. All diagonal entries are therefore equal, so $A = \lambda I_n$. Conversely, for every B, $(\lambda I_n)B = \lambda B = B(\lambda I_n)$. The proof uses only the scalars $0, 1$ and field multiplication, so it applies to finite fields as well.

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EXERCISE 5.15 CHALLENGE

Let $n \geq 2$. Let W consist of all real $n \times n$ matrices whose row sums and column sums all equal the same number. Please prove that W is a subspace and find a basis and its dimension. You may use E_{ij}, the matrix with a single 1 in position (i, j).

WORKED SOLUTION

Addition adds the common sums, scalar multiplication scales them, and zero belongs to W. Hence W is a subspace. Let W_0 be the matrices with all row and column sums zero. For $i, j < n$, define

$$F_{ij} = E_{ij} - E_{in} - E_{nj} + E_{nn}.$$

Each lies in W_0. Every $A \in W_0$ satisfies

$$A = \sum_{i=1}^{n-1} \sum_{j=1}^{n-1} a_{ij} F_{ij}.$$

The upper left $(n-1) \times (n-1)$ blocks agree. Zero row sums then determine the last column, zero column sums determine the last row, and the final entry is determined as well. These matrices are independent: entry (i, j) with $i, j < n$ reads off the coefficient of F_{ij}. If the common sum of $A \in W$ is c, then $A - cI_n \in W_0$. Since $I_n \notin W_0$, a basis is

$$\{I_n\} \cup \{F_{ij} : 1 \leq i, j < n\}, \qquad \dim W = 1 + (n-1)^2.$$

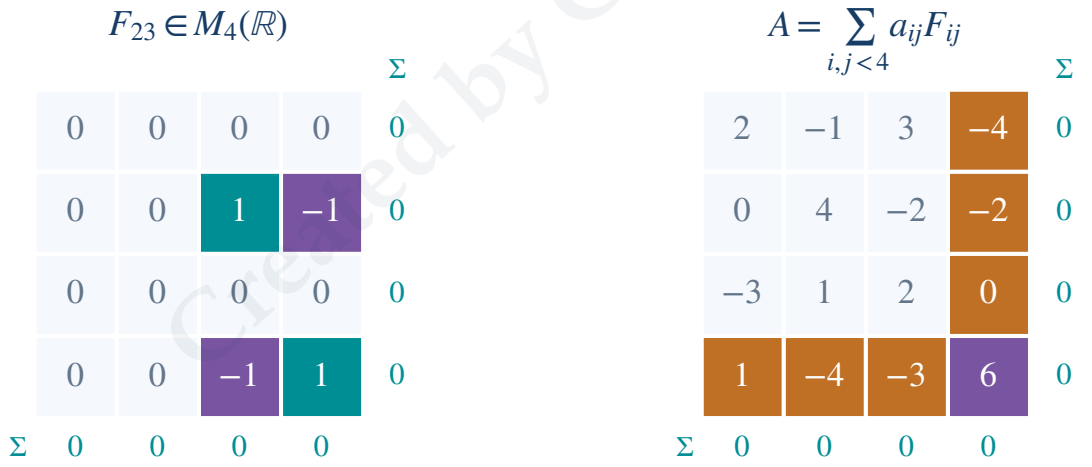


Figure 5.4. Left: the matrix F_{23} for $n = 4$. Right: an interior block completed so that every row and column sum is zero.

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EXERCISE 5.16 CHALLENGE

A permutation matrix has exactly one entry equal to 1 in each row and each column, with all other entries zero. Let $n \geq 2$. Please prove that $A \in M_n(\mathbb{R})$ is a linear combination of permutation matrices if and only if all its row sums and column sums equal the same number.

WORKED SOLUTION

Every permutation matrix has row and column sums equal to 1. Thus every linear combination has all those sums equal to the sum of its coefficients.
Conversely, let c be the common sum of A and put $B = A - cI_n$. All row and column sums of B are zero. For $i, j < n$, define

$$F_{ij} = E_{ij} - E_{in} - E_{nj} + E_{nn},$$

where E_{ij} has one 1 in position (i, j) and zeros elsewhere. Then

$$B = \sum_{i=1}^{n-1} \sum_{j=1}^{n-1} b_{ij} F_{ij}.$$

The upper left blocks agree, and zero row and column sums determine the remaining entries. Each F_{ij} is the difference of two permutation matrices. Choose a permutation matrix P with ones in positions (i, j) and (n, n), matching the remaining $n - 2$ rows and columns in any order. Swap rows i, n to obtain a permutation matrix Q. Then

$$P - Q = E_{ij} + E_{nn} - E_{in} - E_{nj} = F_{ij}.$$

Thus every F_{ij} is a linear combination of permutation matrices. The identity is itself a permutation matrix, so $A = B + cI_n$ is a linear combination of them as well.

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